MACHINE LEARNING PROJECT

Arthritis Prediction

Using BRFSS 2024 Behavioral Health Survey Data

Yashwanth Kotha | Yashwanth Gavini | Roshini Krishnamoorthy Thenmozhi
Data Science | College of Arts & Science | 2026

457K+	**18**	**9**	**0.8024**
Survey Respondents	Features Used	ML Models Tested	Best AUC Score

RESEARCH QUESTION

*“**Which machine learning classifier best predicts arthritis risk from BRFSS behavioral health survey data,** and can hyperparameter tuning or imputation bring us closer to a reliable early detection tool for chronic conditions?”*

Algorithm Selection

9 classifiers benchmarked

Hyperparameter Tuning

RandomizedSearchCV

Data Imputation

Simple Imputation

Dataset: BRFSS 2024 (433K respondents) · 18 behavioral health features · Target: Arthritis diagnosis (binary) · Three experimental conditions: pre-imputation baseline, 100K stratified sample, and full dataset with Simple Imputation

Dataset Overview

BRFSS 2024

Behavioral Risk Factor Surveillance System

Total Respondents:	457,670
Original Columns:	301
Selected Features:	18
Target Variable:	Arthritis (Binary)
File Format:	XPT (SAS Export)
Scenarios Tested:	3 (Pre-imp · 100K · Full)

Selected Features (18)

Health Status	General health, physical health days, mental health days
Conditions	Asthma, coronary heart disease, oral health
Physical	BMI, height, weight, physical activity level
Demographics	Age group, sex, state, urban/rural status
Lifestyle	Tobacco use, alcohol consumption, child count
Prevention	Pneumonia vaccination status

Methodology

01

Data Loading

Chunked SAS XPT reading (10K rows/chunk) to handle 296-column, 457K-row file without memory errors. Selected 18 BRFSS feature columns.

02

Preprocessing

Renamed BRFSS codes to readable labels. Replaced coded missing values (7 = Don't Know, 9 = Refused) with NaN. Scaled Weight and BMI ÷ 100.

03

Missing Analysis

Identified missing data across 19 columns. Pneumonia Vaccination highest at 16.1%. Visualized via heatmap and percentage summary table.

04

3 Scenarios

① Pre-imputation: drop null rows, base models only.
② 100K stratified sample with imputation, base + tuned.
③ Full 433K dataset with imputation.

05

Modeling

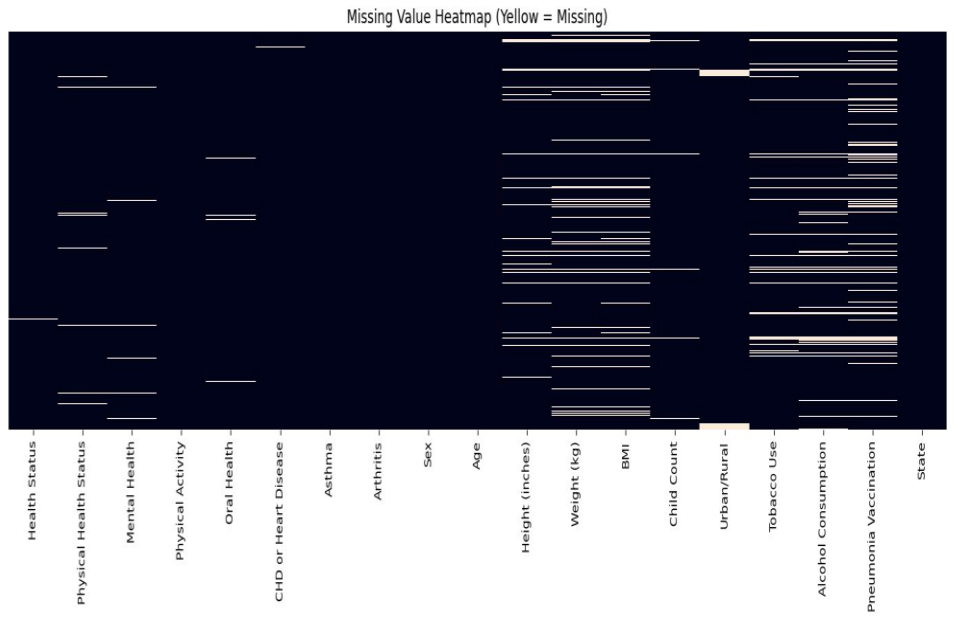
9 classifiers benchmarked: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, AdaBoost, KNN, Naive Bayes, MLP.

06

Tuning & Evaluation

RandomizedSearchCV with StratifiedKFold (5-fold) optimized for ROC-AUC. Metrics reported: Accuracy, AUC, F1-Score, Precision, Recall.

Missing Value Analysis

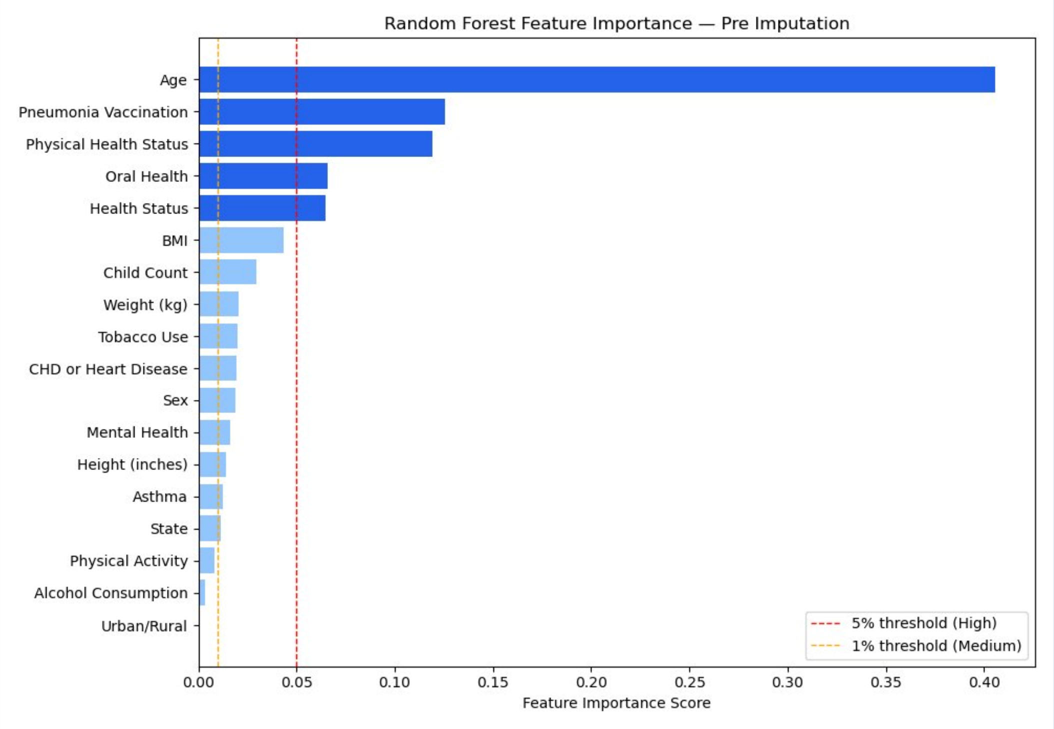


Top Missing Features

Pneumonia Vaccination	73,859	16.1%
Alcohol Consumption	43,777	9.6%
BMI	43,037	9.4%
Weight (kg)	36,392	7.9%
Tobacco Use	32,022	7.0%
Height (inches)	31,838	7.0%
Urban/Rural	14,623	3.2%
Physical Health Status	11,067	2.4%

Strategy: Coded missing values (7 = Don't Know, 9 = Refused) replaced with NaN. Simple imputation applied after ordinal encoding to retain all valid records, preventing the data loss that would come from simply dropping missing rows.

Feature Importance Pre-Imputation (Random Forest)

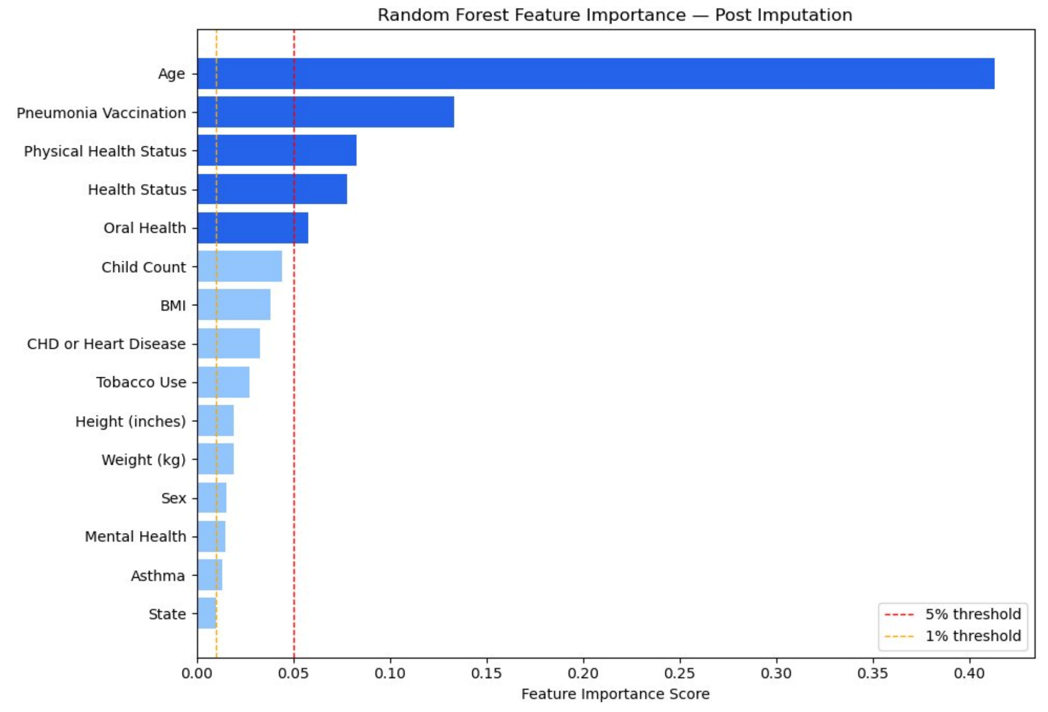


Top Feature Scores

1. Age	0.406
2. Pneumonia Vaccination	0.125
3. Physical Health Status	0.119
4. Oral Health	0.066
5. Health Status	0.065
6. BMI	0.044
7. Child Count	0.030
8. Weight (kg)	0.021
9. Tobacco Use	0.020

Age dominates at 40.6% · Pneumonia Vaccination (12.5%) and Physical Health Status (11.9%) are strong secondary predictors · 17 features present pre-imputation; Alcohol Consumption, Urban/Rural, and Physical Activity drop post-imputation

Feature Importance Post-Imputation (Random Forest)

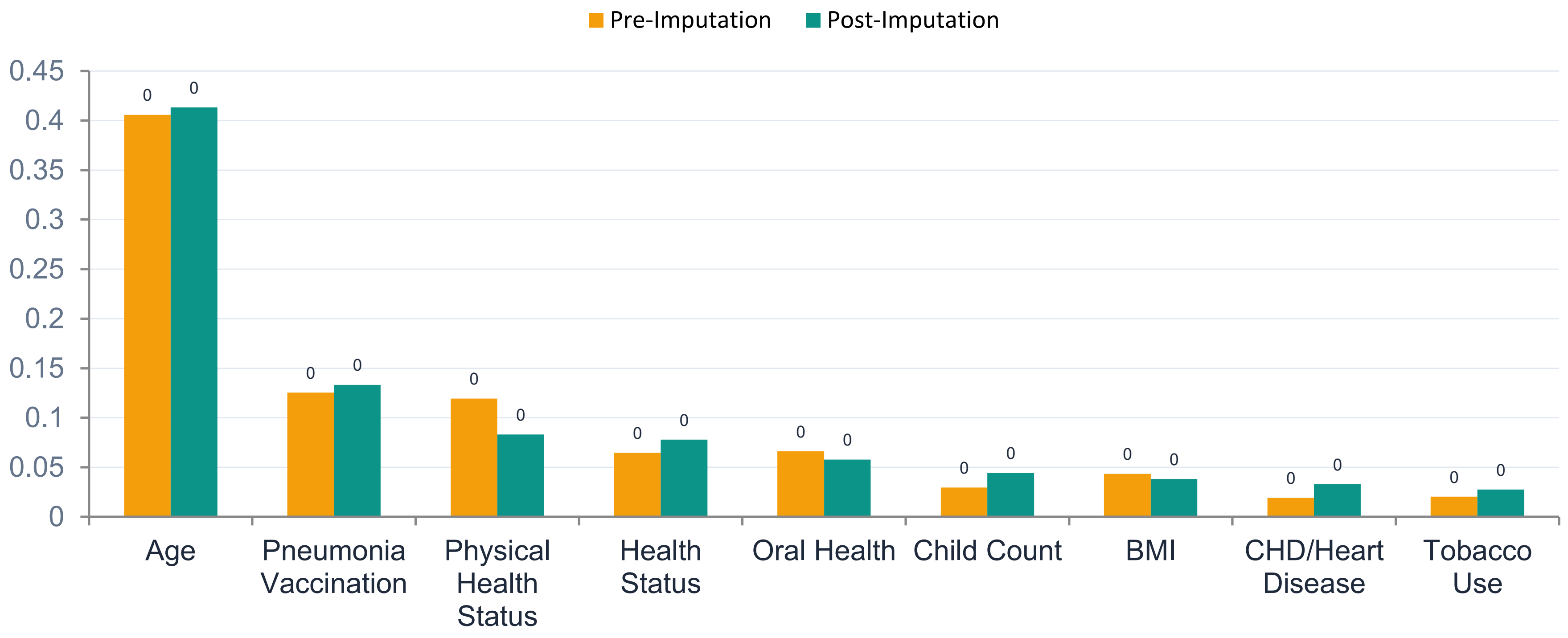


Top Feature Scores

1. Age	0.413
2. Pneumonia Vaccination	0.133
3. Physical Health Status	0.083
4. Health Status	0.078
5. Oral Health	0.058
6. Child Count	0.044
7. BMI	0.038
8. CHD or Heart Disease	0.033
9. Tobacco Use	0.028

Age remains dominant at 41.3% · Health Status rises after imputation (more complete records) · 15 features post-imputation; Alcohol Consumption, Physical Activity, and Urban/Rural removed due to insufficient data density

Feature Importance Pre vs Post Imputation Comparison



Age: +0.008 ↑
Stable dominant predictor

Physical Health Status: −0.036 ↓
Drops significantly after imputation

Health Status: +0.013 ↑
Rises as imputation fills gaps

CHD or Heart Disease: +0.014 ↑
More complete data, stronger signal

Result 1 - Pre-Imputation: Null Rows Dropped, Base Models Only

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	0.7063	0.7384	0.7063	0.7116	0.7942
Decision Tree	0.7020	0.7326	0.7020	0.7074	0.7862
Random Forest	0.7035	0.7397	0.7035	0.7088	0.7947
Gradient Boosting	0.7292	0.7236	0.7292	0.7246	0.7979
XGBoost	**0.7079**	**0.7418**	**0.7079**	**0.7132**	**0.7980**
AdaBoost	0.7244	0.7171	0.7244	0.7131	0.7922
KNN	0.7008	0.6951	0.7008	0.6968	0.7549
Naive Bayes	0.7108	0.7080	0.7108	0.7091	0.7713
MLP	0.7281	0.7234	0.7281	0.7247	0.7961

Best Model (ROC-AUC): XGBoost with AUC = 0.7980 | Null rows dropped · Base models only · No imputation

Result 2 - Post-Imputation: 100K Stratified Sample, Base vs Tuned

Model	Base Acc	Base AUC	Base F1	Tuned Acc	Tuned AUC	Tuned F1	AUC Gain
Logistic Regression	0.7117	0.7987	0.7178	0.7120	0.7987	0.7181	≈ +0.0000
Decision Tree	0.7034	0.7877	0.7096	0.7053	0.7894	0.7114	↑ +0.0017
Random Forest	0.7073	0.7979	0.7135	0.7084	0.7985	0.7146	≈ +0.0005
Gradient Boosting	0.7357	0.8011	0.7315	0.7343	0.8017	0.7306	≈ +0.0006
XGBoost ★	**0.7116**	**0.8013**	**0.7178**	**0.7099**	**0.8020**	**0.7161**	**≈ +0.0007**
AdaBoost	0.7326	0.7955	0.7216	0.7331	0.7994	0.7272	↑ +0.0038
KNN	0.7137	0.7656	0.7094	0.7198	0.7808	0.7152	↑ +0.0152
Naive Bayes	0.7162	0.7760	0.7156	0.7161	0.7761	0.7155	≈ +0.0001
MLP	0.7356	0.8009	0.7316	0.7331	0.8004	0.7317	≈ −0.0005

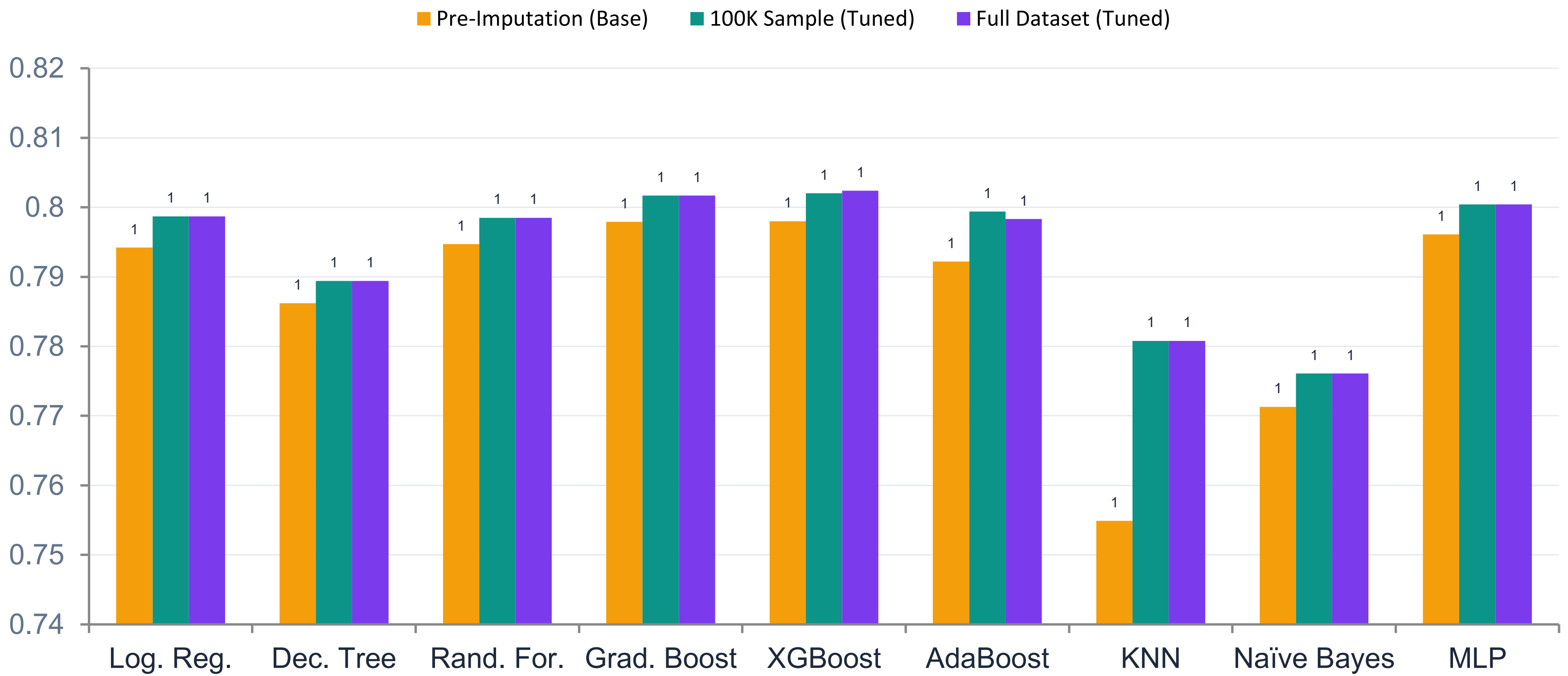
Best Model (Tuned ROC-AUC): XGBoost with AUC = 0.8020 · 100K stratified sample with imputation

Result 3 - Post-Imputation: Full Dataset (433K), Base vs Tuned

Model	Base Acc	Base AUC	Base F1	Tuned Acc	Tuned AUC	Tuned F1	AUC Gain
Logistic Regression	0.7117	0.7987	0.7178	0.7120	0.7987	0.7181	≈ +0.0000
Decision Tree	0.7034	0.7877	0.7096	0.7053	0.7894	0.7114	↑ +0.0017
Random Forest	0.7073	0.7979	0.7135	0.7084	0.7985	0.7146	≈ +0.0005
Gradient Boosting	0.7357	0.8011	0.7315	0.7342	0.8017	0.7305	≈ +0.0006
XGBoost ★	**0.7109**	**0.8015**	**0.7171**	**0.7114**	**0.8024**	**0.7175**	**≈ +0.0009**
AdaBoost	0.7130	0.7814	0.6888	0.7319	0.7983	0.7250	↑ +0.0169
KNN	0.7137	0.7656	0.7094	0.7198	0.7808	0.7152	↑ +0.0152
Naive Bayes	0.7162	0.7760	0.7156	0.7161	0.7761	0.7155	≈ +0.0001
MLP	0.7356	0.8009	0.7316	0.7331	0.8004	0.7317	≈ −0.0005

Best Model (Tuned ROC-AUC): XGBoost — AUC = 0.8024 · Full 433K dataset with KNN imputation

Cross-Scenario AUC Comparison — All 3 Results



Imputation consistently raises AUC across all 9 models · XGBoost leads in every scenario · KNN shows the largest imputation benefit (+0.023)

Best Model - XGBoost Deep Dive

0.8024
ROC-AUC
Full Dataset · Tuned

71.1%
Accuracy
Full Dataset · Tuned

0.7175
F1-Score
Full Dataset · Tuned

+0.0009
AUC Gain
vs Base XGBoost

Model Configuration

n_estimators:	**100**
max_depth:	**6**
learning_rate:	**0.1**
subsample:	**0.8**
colsample_bytree:	**0.8**
scale_pos_weight:	**neg/pos ratio**
eval_metric:	**logloss**

Why XGBoost Won

- Handles class imbalance via scale_pos_weight
- Gradient boosting captures non-linear feature interactions
- L1/L2 regularization prevents overfitting on large data
- Consistent #1 AUC across all 3 experimental scenarios
- Base model near-optimal tuning gains marginal but confirmed

Results

Imputation Preserves Valuable Health Information

Imputation retained valuable health information that would have been lost by simply dropping missing rows, preserving these patterns likely contributed to stronger model predictions across all classifiers.

All 9 Classifiers Exceeded AUC 0.76

XGBoost achieved the best tuned AUC of 0.802 after RandomizedSearchCV optimization, demonstrating that behavioral survey data alone is sufficient to build a strong predictive model, no clinical test results required.

Age, Pneumonia Vaccination & Physical Health Lead

Age, Pneumonia Vaccination, and Physical Health Status consistently ranked as the top predictors, reinforcing that simple, routinely collected health indicators carry significant power in identifying arthritis risk.

CONCLUSION

Conclusion

Machine learning transforms routine health data into a silent guardian, capable of identifying chronic disease risk before it becomes a crisis, because in healthcare, the most powerful intervention is the one that arrives before it is too late.

This study proves that affordable, routinely collected health data, combined with the right predictive pipeline, can serve as a scalable early-warning system, not just for arthritis, but across the broader landscape of chronic conditions where catching the signal early can mean the difference between prevention and a lifetime of treatment.

0.802
Best AUC — XGBoost

9
Classifiers Benchmarked

433K
Survey Respondents

41.3%
Age Feature Importance